

SmartStudy: A Lightweight Machine Learning System for Student Performance Prediction

Abstract

This paper presents SmartStudy, a lightweight machine learning system designed to predict student academic performance using attendance, study hours, assignment scores, and previous exam results. The system aims to help educators identify students who may need additional academic support.

1. Introduction

Educational institutions collect many indicators related to student performance. However, teachers often have limited time to analyze these indicators manually. SmartStudy addresses this problem by combining commonly available student records into a predictive model.

2. Problem Statement

The main problem is the early identification of students who are at risk of obtaining low final examination scores. The proposed system attempts to provide an automated prediction using historical academic and engagement data.

3. Methodology

The dataset contains records from 1,200 students. Four input features are used: attendance percentage, weekly study hours, assignment average, and previous examination score. The data is divided into 80% training data and 20% testing data. A Random Forest classifier is trained to classify students into three categories: low, medium, and high expected performance.

4. Dataset

The dataset contains 1,200 student records collected over one academic year. Attendance ranges from 45% to 100%, study time ranges from 2 to 30 hours per week, and assignment scores are recorded on a 0 to 100 scale. Previous examination scores are also represented on a 0 to 100 scale.

5. Results

The Random Forest model achieved an accuracy of 87.5% on the test set. Attendance and previous examination score were the two most influential features. The results indicate that combining multiple academic indicators can improve early identification of students requiring support.

6. Limitations

The study uses data from a single academic institution, so the results may not generalize to other institutions. The dataset also does not include socioeconomic factors, learning disabilities, or qualitative teacher assessments. In addition, the model predicts performance categories rather than exact final grades.

7. Future Work

Future research will evaluate the system on datasets from multiple institutions and investigate additional features such as learning activity, course difficulty, and student feedback. The authors also plan to compare Random Forest with gradient boosting and neural network models.

8. Conclusion

SmartStudy demonstrates that a lightweight machine learning approach can predict student performance from commonly available academic indicators. The proposed system can support educators by identifying students who may benefit from additional assistance.